Collisions Transfer Energy

Chapter 5 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION How is energy involved in collisions?

KEY WORDS

collide — To collide is when two objects _____ with impact (make contact).

collision — A collision happens when two objects _____ with impact.

MODEL IT

In a collision, a moving object's energy of motion can transfer to the other object. Energy does NOT disappear.

COLLISION

↓

MOTION CAN CHANGE

↓

ENERGY CAN TRANSFER

↓

ENERGY DOES NOT DISAPPEAR

DRAW / LABEL — BEFORE · DURING · AFTER

Draw the two cars BEFORE, DURING, and AFTER they collide. Use arrows to show motion.

BEFORE	DURING	AFTER

Evidence + Fair Test

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WHAT CHANGES IN A COLLISION?

List the changes a collision can cause, with an example from our Crash Lab.

CHANGE	EXAMPLE FROM OUR CRASH LAB
motion (speed or direction)	
shape	
sound	

FAIR TEST

A fair test changes ONE thing at a time. Compare the SAME object at different speeds.

Part A — the ONE thing we changed was _____ (the SAME car).

CRASH LAB EVIDENCE

SAME CAR, RAMP HEIGHT	TARGET DISPLACEMENT (CM)
low (slower)	
high (faster)	

The faster the SAME car moved, the _____ the target moved. That change is evidence of MORE energy of motion.

CHECK YOUR THINKING

After a crash, the cars stop. Did the energy disappear? Explain using motion and sound.

CONNECT *Chapter 1 → 5: energy causes change; a collision transfers energy and makes sound.*